Quantum Denoising for Brain-Computer Interfaces

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Introduction

Brain-Computer Interfaces (BCIs) enable direct communication between the brain and external devices, with electroencephalography (EEG) being a key modality due to its non-invasive nature. However, EEG signals are often noisy, affected by muscle activity, eye movements, and environmental interference, which limits BCI performance. This study investigates Quantum Variational Autoencoders (QVAEs) for EEG denoising, leveraging quantum computing principles to enhance signal quality.

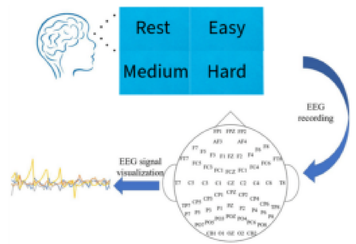


Figure 1. EEG acquisition process with different difficulty levels [3].

Classical vs. Quantum Autoencoders

Classical Autoencoders (CAE): CAEs are neural networks designed for unsupervised learning. They consist of an encoder that compresses input data into a latent space and a decoder that reconstructs the original data:

$$L = \|x - \hat{x}\|^2$$

While effective for many applications, CAEs struggle with non-linear, high-dimensional noise typical in EEG signals.

Quantum Autoencoders (QAE): QAEs extend CAEs to the quantum realm using parameterized quantum circuits. They aim to compress quantum states, preserving essential information while discarding noise:

$$U(\theta)|\psi\rangle_{AB} \approx |\phi\rangle_A \otimes |0\rangle_B$$

Fidelity is measured via the SWAP test:

$$F = 1 - 2P(|1\rangle)$$

where $P(|1\rangle) = \frac{L}{M}$ is the probability of measuring the auxiliary qubit in the $|1\rangle$ state, with L and M representing the number of outcomes in $|1\rangle$ state and the number of trials respectively.

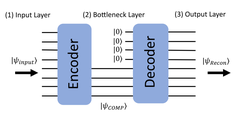


Figure 2. Pictorial Representation of a Quantum Autoencoder [4].

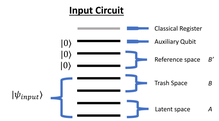


Figure 3. Pictorial Representation of input state of Quantum Autoencoder [4].

Methodology

Dataset: EEG data from cognitive tasks with varying difficulty levels were segmented into 2-second epochs, filtered (1–35 Hz), and downsampled to 128 Hz [2].

Preprocessing: Independent Component Analysis (ICA) was used for artifact removal. Simple Covariance matrices were computed:

$$C_i = \frac{1}{T-1} X_i X_i^T$$

where X_i represents EEG data for the ith epoch.

Feature Extraction: Covariance matrices were projected to the tangent space of the Riemannian manifold [1].

Quantum Denoising: The preprocessed data were input to a Quantum Variational Autoencoder (QVAE) with a 9-qubit RealAmplitudes ansatz. The SWAP test was used for fidelity evaluation.

Evaluation: QVAE performance is assessed using reconstruction loss and classification accuracy via LDA with 5-fold cross-validation, measured by ROC AUC scores. Feature extraction and quantum processing are implemented using **pyriemann-qiskit**, which is a framework adapted to the processing and classification of EEG signals using Riemannian geometry and quantum algorithms.

Riemannian Geometry for EEG Analysis

EEG covariance matrices lie on a Riemannian manifold of Symmetric Positive Definite (SPD) matrices rather than in Euclidean space. Standard vector-based operations fail to preserve the geometric relationships between these matrices, making Riemannian approaches more suitable for feature extraction.

Why Riemannian Geometry? Since SPD matrices belong to a curved manifold, distances between them must be computed using geodesics rather than Euclidean norms. The geodesic distance between two covariance matrices C_1 and C_2 is:

$$\delta_R(C_1, C_2) = \left\| \log(C_1^{-1} C_2) \right\|_F$$

where $\|\cdot\|_F$ is the Frobenius norm. This ensures that distances respect the manifold's curvature.

Tangent Space Mapping: To apply standard classification techniques, covariance matrices are projected onto a tangent space at a reference point P using the logarithm map:

$$S_i = \log_P(C_i) = P^{1/2} \log(P^{-1/2} C_i P^{-1/2}) P^{1/2}$$

where P is the geometric mean of all covariance matrices. This transformation provides a vectorized representation while retaining the manifold's intrinsic structure [1].

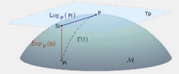


Figure 4. Tangent space of $\mathcal{M}$ at $\mathbf{P}$, with $\mathbf{S_i}$ as the tangent vector and $\gamma(t)$ the geodesic to $\mathbf{P_i}$ [1].

Advantages in EEG Analysis

- Is computationally inexpensive (as compared to e.g., deep learning)
- Enables geodesic-based distance computations for robust classification.
- Enhances generalization across subjects and experimental sessions.

By leveraging Riemannian geometry, EEG feature extraction becomes more stable and reliable, leading to improved classification performance in BCIs.

Results

Performance Metrics: ROC AUC scores compared classical LDA and QNN+LDA pipelines.

Pipeline	AUC Score	Execution Time (s)
LDA	0.7177	0.42
QNN+LDA	0.6686	2971.01

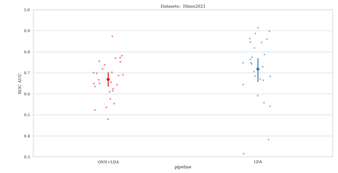


Figure 5. ROC AUC comparison between classical LDA and QNN+LDA pipelines.

Quantum denoising showed enhanced robustness but slightly lower AUC scores compared to classical methods, with increased computational costs.

Discussion and Future Work

Key Findings: This study demonstrates the feasibility of Quantum Variational Autoencoders (QVAEs) for EEG denoising in passive Brain-Computer Interfaces (pBCIs). While QVAEs offer stability and noise reduction, their classification performance remains slightly lower than classical methods like LDA, with increased computational costs.

Challenges

- High computational cost of quantum circuit simulations.
- Sensitivity to hyperparameter tuning, particularly latent space size.
- Limited generalization across subjects and experimental sessions.

Future Directions

- Optimizing quantum circuit architectures for improved efficiency.
- Developing hybrid quantum-classical models for feature extraction.
- Exploring near-term quantum hardware implementations for practical use.

References

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[4] QiskitMachineLearning. The Quantum Autoencoder - Qiskit Machine Learning 0.8.1 — qiskit-community.github.io. `https://qiskit-community.github.io/qiskit-machine-learning/tutorials/12_quantum_autoencoder.html`, 2024. [Accessed 13-02-2025].